



Software Project Management-506

Semester 6th

Name: _____

Roll Number: _____

Computer Science Department
University Of Karachi

University Of Karachi



Course Name (Course Code) _____ Semester _____ Batch _____
 Name of Student: _____ Roll No. _____

Lab	Description & Score				
1.	Software Handling ()/2	Programming/ Simulations ()/5	Results ()/2	Lab Report ()/1	Score ()/10
2.	Software Handling ()/2	Programming/ Simulations ()/5	Results ()/2	Lab Report ()/1	Score ()/10
3.	Software Handling ()/2	Programming/ Simulations ()/5	Results ()/2	Lab Report ()/1	Score ()/10
4.	Software Handling ()/2	Programming/ Simulations ()/5	Results ()/2	Lab Report ()/1	Score ()/10
5.	Software Handling ()/2	Programming/ Simulations ()/5	Results ()/2	Lab Report ()/1	Score ()/10
6.	Software Handling ()/2	Programming/ Simulations ()/5	Results ()/2	Lab Report ()/1	Score ()/10
7.	Software Handling ()/2	Programming/ Simulations ()/5	Results ()/2	Lab Report ()/1	Score ()/10
8.	Software Handling ()/2	Programming/ Simulations ()/5	Results ()/2	Lab Report ()/1	Score ()/10
9.	Software Handling ()/2	Programming/ Simulations ()/5	Results ()/2	Lab Report ()/1	Score ()/10
10.	Software Handling ()/2	Programming/ Simulations ()/5	Results ()/2	Lab Report ()/1	Score ()/10
11.	Software Handling ()/2	Programming/ Simulations ()/5	Results ()/2	Lab Report ()/1	Score ()/10
12.	Software Handling ()/2	Programming/ Simulations ()/5	Results ()/2	Lab Report ()/1	Score ()/10

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		()/5			
13.	<i>Software Handling</i> ()/2	<i>Programming/ Simulations</i> ()/5	<i>Results</i> ()/2	<i>Lab Report</i> ()/1	<i>Score</i> ()/10
14.	<i>Software Handling</i> ()/2	<i>Programming/ Simulations</i> ()/5	<i>Results</i> ()/2	<i>Lab Report</i> ()/1	<i>Score</i> ()/10
TOTAL SCORE = 140				OBTAINED SCORE	

Overall Score: _____ out of 50

Examined by: _____

Overall Formula= (Obtained Score / Total Score) x 50 (Name and Signature of Concerned)



Software Project Management-LAB (3+0)

Instructor: Dr. Humera

Office Hours: ()

Prerequisite: Programming Fundamentals, Software Engineering, Data Structures and Algorithms, Object-Oriented Programming (OOP), Database Management Systems (DBMS)

Objectives:

The primary objective of the Software Project Management lab is to provide students with hands-on experience in applying project management concepts and tools to real-world software development scenarios. Through practical exercises, students will develop the skills necessary to plan, execute, monitor, and control software projects effectively.

Contents:

- ❖ Focus on practical application of project management concepts.
- ❖ Cover project planning, execution, monitoring, and closure phases.
- ❖ Utilize project management software and collaboration tools.
- ❖ Employ case studies, simulations, and group projects for hands-on learning.

Learning Outcomes:

Mapping to Course Learning Outcomes (CLOs) and Program Learning Outcomes (PLOs)

LO	CLO	PLO
LO1	CLO1: Define project management concepts and processes	PLO1: Apply fundamental knowledge of computer science and software engineering
LO2	CLO2: Analyze project requirements and scope	PLO2: Analyze and solve complex problems
LO3	CLO3: Develop project plans and documentation	PLO3: Design and implement software solutions
LO4	CLO4: Work effectively in teams	PLO4: Communicate effectively
LO5	CLO5: Present technical information clearly and concisely	PLO5: Communicate effectively



Recommended Book:

- ❖ Software Project Management Bob Hughes and Mike Cotterell, McGraw-Hill
- ❖ Software Project Management in Practice Peter Gordon, Pankaj Jalote
- ❖ Quality Software Project Management Futrell, Robert T., Donald F. Shafer, and Linda I. Shafer

Reference Books:

1. A Guide to the Project Management Body of Knowledge, 5th Edition (PMBOK Guides),
2. The Mythical Man-Month: Essays on Software Engineering, by Frederick P. Brooks.
3. Mastering Software Project Management: Best Practices, Tools and Techniques, Murali K. Chemuturi and Thomas M. Cagley Jr., J. Ross Publishing, 2010
4. Effective Project Management: Traditional, Agile, Extreme, Robert K. Wysocki, Wiley; 6th Edition, 2011

Administrative Instructions:

- According to institute policy, 75% attendance is *mandatory* to appear in the final examination but 100% will be expected. Approved leaves will not be considered towards attendance.
- Every student should bring book and manual in each lab.
- Every student is expected to be in lab before schedule starting time.
- In any case there will be no rescheduling and makeup of labs.



Assignment-01

Project Charter Development

Objectives:

The purpose of this lab is to get Students familiar for forming teams and develop a project charter for a hypothetical software project. The project charter will include a project title, objectives, stakeholders, deliverables, and a timeline. I hope this design lab activity helps students to learn about project charter development.

Apparatus:

Hardware Requirement

Personal computer, **Laptops or tablets:** For accessing project management tools or templates

Software Requirement

Internet access: For research and collaboration.

THEORY:.

What is a Project Charter?

A project charter is a formal document that authorizes the start of a project. It outlines the project's objectives, scope, stakeholders, and high-level requirements. Essentially, it's the project's initial blueprint, providing direction and alignment for the entire team.

Example: A software development company wants to develop a new mobile app. The project charter for this initiative would outline the app's purpose (e.g., to increase customer engagement), target audience, key features, and the project manager responsible.

Purpose and Importance of a Project Charter

The project charter serves as a foundational document that:

- **Authorizes the project:** Gives formal approval to proceed with the project.
- **Defines project scope:** Clearly outlines the project's boundaries, preventing scope creep.
- **Aligns stakeholders:** Ensures everyone understands the project's goals and objectives.
- **Establishes project governance:** Defines roles, responsibilities, and decision-making authority.
- **Manages expectations:** Sets clear expectations for project outcomes and deliverables.

Example: A construction company plans to build a new office building. The project charter would outline the project's purpose (e.g., accommodate growing workforce), scope (building size, number of floors), key stakeholders (investors, architects, contractors), and the project manager's authority to make decisions.



Key Components of a Project Charter

A comprehensive project charter typically includes:

- **Project title and identification:** A clear and concise project name.
- **Project initiation:** Background, reason for the project, and business case.
- **Project objectives:** Specific, measurable, achievable, relevant, and time-bound (SMART) goals.
- **Project scope:** A high-level overview of project deliverables and boundaries.
- **Project stakeholders:** Identification of key individuals or groups with an interest in the project.
- **Project assumptions and constraints:** Factors that may influence the project.
- **Project schedule and budget:** Preliminary estimates for project timeline and resources.
- **Project manager and authority levels:** Designation of the project manager and their authority.
- **Project approvals:** Signatures from key stakeholders to authorize the project.

Example: A marketing team is launching a new product. The project charter would outline the product name, target market, key marketing channels, budget allocation, and the project manager responsible for the campaign.

Lab #01 task:

Activities:

1. **Form teams:** Students will form teams of 4-5 people.
2. **Brainstorm project ideas:** Teams will brainstorm a list of potential software projects.
3. **Select a project:** Teams will select one project to focus on.
4. **Develop a project charter:** Teams will develop a project charter for their selected project. The project charter should include the following:
 - Project title
 - Project objectives
 - Project stakeholders
 - Project deliverables
 - Project timeline
5. **Present project charters:** Teams will present their project charters to the class in the form of presentation.

**NOTE:**

- Students should be creative and innovative in their project ideas.
- Students should identify the key elements of a project charter.

Assessment:

Students will be assessed on their ability to:

- Form a team and work collaboratively
- Brainstorm and select a project
- Develop a clear and concise project charter
- Present their project charter effectively

Additional Resources:

- Project Charter Template: <https://www.projectmanagement.com/project-charter-template>
- Project Management Body of Knowledge (PMBOK Guide): <https://www.pmi.org/about/standards-certification/pmbok-guide>
- Agile Project Management: <https://www.agilealliance.org/>